

Quantum Compressed Sensing Assisted LSTM-Based Pilot Contamination Mitigation for Massive MIMO-FBMC Systems

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Abstract – The combination of Filter Bank Multi-Carrier (FBMC) and Massive Multiple-Input Multiple-Output (Massive MIMO) has emerged as a very promising wireless communication architecture for the upcoming 6G networks, mainly because of its reduced out-of-band emissions and improved spectral efficiency. However, pilot contamination is a major problem that makes accurate channel estimation difficult and negatively impacts system performance in multi-cell settings. In order to address this issue, this work presents a Quantum Compressed Sensing Assisted Long Short-Term Memory (QCS-LSTM) framework designed to reduce pilot contamination and enable reliable channel estimation in Massive MIMO-FBMC systems. The proposed methodology uses channel sparsity through the principles of compressed sensing and combines quantum-inspired encoding and measurement techniques to enhance the efficiency of sparse signal recovery. An LSTM network is then employed to capture temporal fluctuations in the channel and lessen the effect of pilot interference during the estimation phase. This integrated design allows for better channel state information extraction, which reduces estimation errors even in the case of pilot contamination. Performance is assessed in different Signal-to-Noise Ratio (SNR) scenarios using metrics such as Bit Error Rate (BER), Mean Square Error (MSE), channel recovery accuracy, and spectral efficiency. Simulation findings indicate that the proposed QCS-LSTM model exhibits superior estimation accuracy and contamination resistance compared to traditional Least Squares (LS), compressed sensing, and independent deep learning methodologies. Furthermore, the framework exhibits improved robustness for dynamic wireless environments and supports scalable deployment for next-generation intelligent communication systems. The proposed methodology provides a potential pathway toward efficient AI-enabled and quantum-enhanced wireless channel estimation for future Massive MIMO-FBMC networks.

Keywords- Quantum Compressed Sensing (QCS), Long Short-Term Memory (LSTM), Pilot Contamination, Massive MIMO, FBMC, Channel Estimation, Sparse Signal Recovery, 6G Networks.

I. INTRODUCTION

Future wireless communication networks are expected to provide extremely high data rates, massive connectivity, intelligent resource allocation, and ultra-low latency services to support emerging applications in sixth-generation (6G) environments [1]. Massive Multiple-Input Multiple-Output (Massive MIMO) has emerged as one of the most important enabling technologies because of its capability to improve spectral efficiency, transmission reliability, and network capacity through the deployment of a large number of antennas at the base station [1], [2]. Along with Massive MIMO, Filter Bank Multi-Carrier (FBMC) modulation has received considerable attention due to its excellent frequency localization characteristics and lower out-of-band emissions compared with conventional Orthogonal Frequency Division Multiplexing (OFDM) systems [3]. However, the challenge of efficient channel estimation remains a significant barrier to the practical deployment of Massive MIMO-FBMC systems, notwithstanding the considerable benefits they offer. Another important aspect which affects the performance of the estimation is the pilot contamination. This is caused by the reuse of the same pilot sequences in the neighbouring cells which causes interference in the channel acquisition process [2] and [4]. This results in inaccurate Channel State Information (CSI), which in turn deteriorates the accuracy of beamforming, increases the detection errors, and limits the overall throughput of the system. Because of their simplicity and analytical derivation, conventional channel estimate methods like Least Squares (LS) and Minimum Mean Square Error (MMSE) estimation have been extensively used in wireless communication systems [5]. However, in dynamic wireless environments with significant pilot contamination and in densely crowded network scenarios, these approaches frequently see a reduction in performance. Furthermore, these estimating techniques usually require additional processing resources and statistical assumptions.

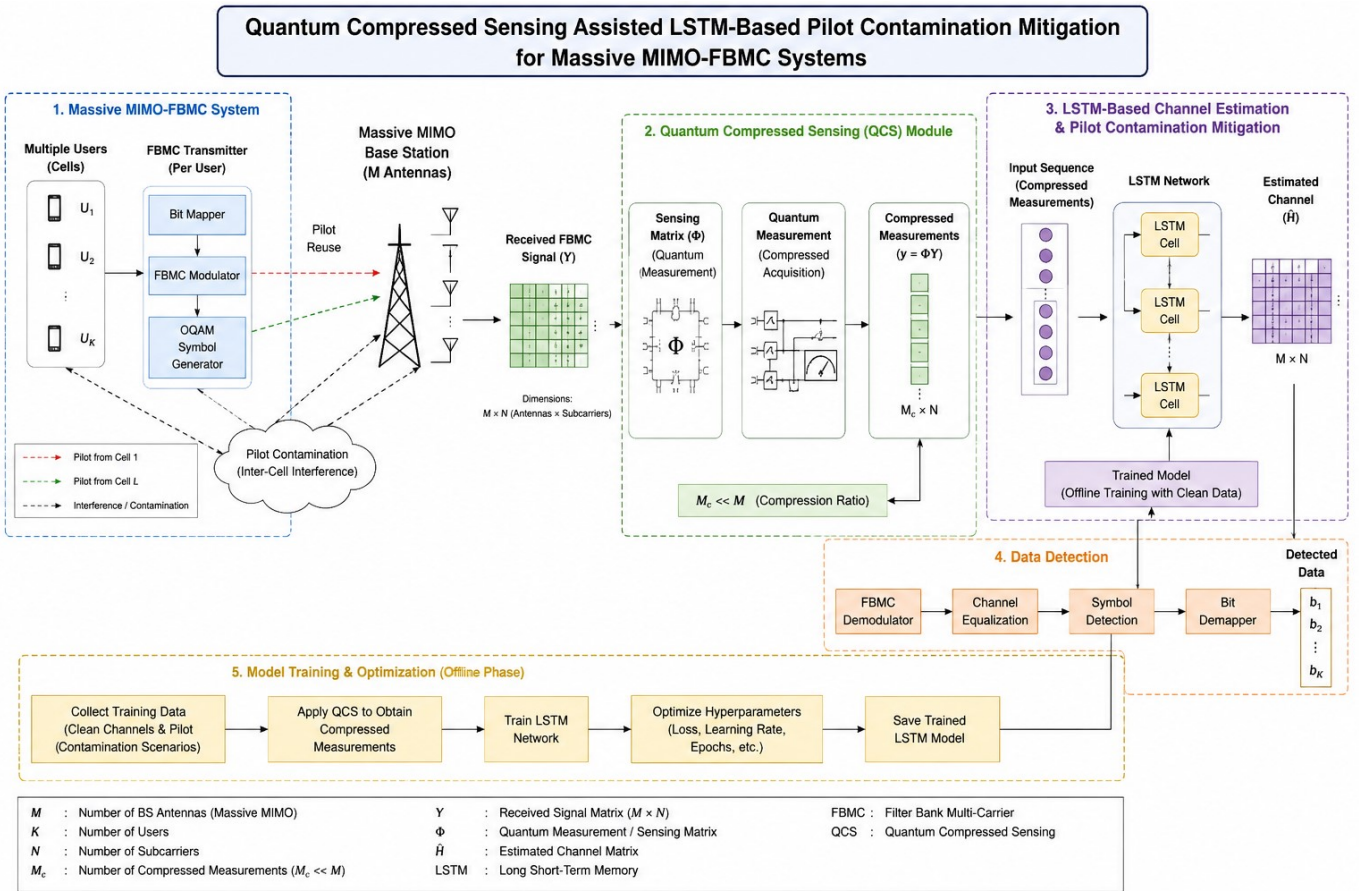


Figure 1: Block diagram of the proposed Quantum Compressed Sensing Assisted LSTM (QCS-LSTM) framework for pilot-contamination mitigation in Massive MIMO-FBMC systems.

Because propagation channels in transformed signal domains are sparse, compressed sensing has become a popular method for wireless channel estimation [6], [7]. Sparse recovery techniques including Orthogonal Matching Pursuit (OMP), Sparse Bayesian Learning (SBL), and Generalized Approximate Message Passing (GAMP) have demonstrated improved channel reconstruction capabilities while reducing pilot overhead. However, in situations with highly contaminated pilot signals and in quickly changing wireless settings, conventional sparse recovery techniques may still encounter difficulties.

New opportunities for improving adaptive channel estimates and wireless communication have been made possible by recent advances in artificial intelligence (AI) [8]. Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) networks are examples of deep learning frameworks that have shown great promise in directly learning complex nonlinear channel parameters from the received data [8]–[10]. While LSTM networks are superior at capturing temporal dependencies and the sequential behavior of signals, CNN

architectures are skilled at extracting spatial aspects of the channel [15], [16]. AI-driven methods for reducing pilot contamination and intelligently assigning pilots in Massive MIMO networks have been the subject of numerous recent studies [11]–[14].

Deep learning-based channel estimation techniques have demonstrated significant improvements in accuracy and robustness compared to conventional algorithms [9], [12], and pilot scheduling and adaptive interference management strategies using reinforcement learning have produced encouraging results in reducing the impact of pilot reuse [13], [14]. Nevertheless, most existing methods focus on either sparse recovery or deep learning separately, failing to take advantage of both temporal and sparse channel characteristics in a cohesive way.

Recently, quantum-inspired computing techniques have attracted increasing attention for communication and signal processing applications [18], [19]. Quantum Compressed Sensing (QCS) extends classical compressed sensing principles by introducing quantum-inspired measurement and sparse reconstruction mechanisms to improve recovery

efficiency and estimation robustness [18]. Combining quantum-inspired sparse recovery with sequential learning architectures offers the possibility of improving channel estimation under severe pilot contamination conditions.

Inspired by these findings, this paper presents a Quantum Compressed Sensing Assisted LSTM-Based Pilot Contamination Mitigation framework designed specifically for Massive MIMO-FBMC systems. This novel approach blends quantum-inspired sparse signal recovery approaches with LSTM-based temporal feature extraction to improve channel estimation performance in pilot-contaminated situations. The framework's performance is measured using measures such as Bit Error Rate (BER), Mean Square Error (MSE), channel recovery accuracy, and spectral efficiency across a variety of Signal-to-Noise Ratio (SNR) circumstances. Simulation findings show that the suggested method greatly enhances estimation capabilities, reduces the effects of pilot contamination, and provides more robustness when compared to traditional estimation methods and AI-assisted benchmarks [10, 20]

Previous research has focused on sparse channel estimation, deep learning-based pilot contamination mitigation, and adaptive pilot scheduling algorithms for Massive MIMO communication systems [11], [12]. While compressed sensing approaches have improved sparse recovery efficiency, they have been demonstrated to be susceptible to extreme contamination circumstances [6], [7]. Deep learning methods including CNN and LSTM achieved higher estimation accuracy by exploiting channel features and temporal behavior [8], [9], [15]. More recently, quantum-inspired signal processing methods have introduced additional opportunities for efficient sparse estimation and intelligent wireless system design [18], [19]. However, integration of Quantum Compressed Sensing with LSTM-based pilot contamination mitigation remains relatively unexplored.

The remainder of this paper is organized as follows. Section II presents the system architecture and mathematical model of the Massive MIMO-FBMC framework. Section III explains the proposed Quantum Compressed Sensing and LSTM-based pilot contamination mitigation methodology. Section IV describes dataset generation, training procedures, and simulation settings. Section V discusses performance evaluation and comparative analysis using BER, MSE, and recovery accuracy metrics. Finally, Section VI concludes the paper and outlines future research directions for quantum-enhanced intelligent wireless communication systems.

II. SYSTEM MODEL

Conventional Massive MIMO systems rely heavily on Orthogonal Frequency Division Multiplexing (OFDM). However, OFDM's rectangular pulsing window causes high out-of-band (OOB) radiation, requiring a Cyclic Prefix (CP) that consumes up to 25% of spectral resources.

Filter Bank Multicarrier (FBMC) eliminates the CP by utilizing a prototype filter bank that isolates sub channels tightly in both the time and frequency domains.

To circumvent Balian-Low theorem constraints (which state that a non-segmented pulse cannot achieve maximum symbol rate and perfect complex orthogonality simultaneously), FBMC shifts to Offset Quadrature Amplitude Modulation (OQAM). A complex QAM symbol $c_{k,n}[m]$ for user k on subcarrier n at time index m is split into its real and imaginary components, staggered by a half-symbol duration $T/2$:

$$c_{k,n}[m] = \Re\{c_{k,n}[m]\} + j\Im\{c_{k,n}[m]\} \quad (1)$$

These are mapped to real-valued symbols $a_{k,n}[m]$ at rate $2/T$:

$$\begin{aligned} a_{k,n}[2m] &= \Re\{c_{k,n}[m]\}, \\ a_{k,n}[2m+1] &= \Im\{c_{k,n}[m]\} \end{aligned} \quad (2)$$

he composite transmitted baseband signal for the k -th user in the l -th cell is formulated as:

$$s_{l,k}(t) = \sum_{m=-\infty}^{\infty} \sum_{n=0}^{N-1} a_{l,k}[m,n] g_{m,n}(t) \quad (3)$$

Where the space-time-frequency localized synthesis basis function $g_{m,n}(t)$ is defined using the prototype filter $g(t)$ (such as the PHYDYAS filter with overlapping factor $K_o = 4$):

$$g_{m,n}(t) = g\left(t - m\frac{T}{2}\right) \exp\left(j2\pi nF\left(t - m\frac{T}{2}\right)\right) \exp(j\phi_{m,n}) \quad (4)$$

The phase term $\phi_{m,n}$ ensures quadrature orthogonality between adjacent subcarriers and time slots:

$$\phi_{m,n} = \frac{\pi}{2}(m+n) + \phi_0 \quad (5)$$

The prototype filter is structurally engineered to satisfy the real-domain orthogonality condition:

$$\Re\left\{\int_{-\infty}^{\infty} g_{m,n}(t) g_{m',n'}^*(t) dt\right\} = \delta_{m,m'} \delta_{n,n'} \quad (6)$$

Where $\delta_{i,j}$ represents the Kronecker delta function. When passed through a time-dispersive fading channel, the orthogonality in the imaginary domain is violated.

The received signal before channel equalization contains an intrinsic interference term:

$$\int_{-\infty}^{\infty} g_{m',n'}(t) g_{m,n}^*(t) dt = j\omega_{m',n'}^{m,n} \quad (7)$$

Where $\omega_{m',n'}^{m,n}$ is a real-valued interference weight. This implies that even under perfect noiseless conditions, the received symbol is corrupted by a purely imaginary crosstalk from its immediate time-frequency neighborhood $\Omega_{m,n}$:

$$Y[m, n] = H[m, n]a[m, n] + \sum_{(m', n') \in \Omega_{m, n}} H[m', n']a[m', n']\omega_{m, n}^{m', n'} + V[m, n] \quad (8)$$

In a multi-cell Massive MIMO infrastructure with L cells, let the target Base Station be located at the epicenter of cell $l=1$. The BS hosts a Uniform Linear Array (ULA) consisting of M elements spaced at $d = \lambda / 2$.

Notations: $\mathbf{x}_t \rightarrow$ input vector at time t , $\mathbf{h}_t^{(1)} \rightarrow$ hidden state of first LSTM layer, $\mathbf{h}_t^{(2)} \rightarrow$ hidden state of second LSTM layer, $\mathbf{C}_t^{(1)}, \mathbf{C}_t^{(2)} \rightarrow$ cell states, $\mathbf{f}_t \rightarrow$ forget gate, $\mathbf{i}_t \rightarrow$ input gate, $\mathbf{o}_t \rightarrow$ output gate, $\mathbf{W} \rightarrow$ input-weight matrices, $\mathbf{U} \rightarrow$ recurrent-weight matrices, $\mathbf{b} \rightarrow$ bias vectors, $\sigma(\cdot) \rightarrow$ sigmoid activation function, $\tanh(\cdot) \rightarrow$ hyperbolic tangent activation function, $\odot \rightarrow$ element-wise multiplication, $\mathbf{N} \rightarrow$ number of complex channel coefficients, $N_{tr} \rightarrow$ number of training sequences, $N_{test} \rightarrow$ number of test sequences.

A. Spatial Channel Vector Model

The physical channel vector $\mathbf{h}_{1,j,k} \in \mathbb{C}^{M \times 1}$ mapping user k in cell j to the target BS in cell 1 is governed by a multipath scattering geometric model:

$$\mathbf{h}_{1,j,k} = \sum_{p=1}^P \beta_{1,j,k}^{(p)} \mathbf{a}(\theta_{1,j,k}^{(p)}) \quad (9)$$

Where $\beta_{1,j,k}^{(p)}$ the complex path is gain of the p -th scatterer, and $\mathbf{a}(\theta)$ is the spatial steering vector defined for an angle of arrival (AoA) θ :

$$\mathbf{a}(\theta) = \left[1, e^{-j2\pi \frac{d}{\lambda} \sin(\theta)}, e^{-j2\pi \frac{2d}{\lambda} \sin(\theta)}, \dots, e^{-j2\pi \frac{(M-1)d}{\lambda} \sin(\theta)} \right]^T \quad (10)$$

During the pilot training interval, all cells reuse an identical set of K mutually orthogonal pilot sequences

$\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_K]^T \in \mathbb{C}^{K \times \tau}$, where $\mathbf{P}\mathbf{P}^H = \tau \mathbf{I}_K$.

The matrix received by the target BS is:

$$\mathbf{Y}_1 = \sum_{k=1}^K \mathbf{h}_{1,k} \mathbf{p}_k^T + \sum_{j=2}^L \sum_{k=1}^K \mathbf{h}_{1,j,k} \mathbf{p}_k^T + \mathbf{J}_{FBMC} + \mathbf{V}_1 \quad (11)$$

Here, $\mathbf{J}_{FBMC}$ represents the structured matrix of the intrinsic imaginary interference generated across the pilot grid blocks. Correlating the raw received matrix with user k 's specific pilot sequence $\mathbf{p}_k^*$ yields:

$$\hat{\mathbf{h}}_{1,k}^{LS} = \frac{1}{\tau} \mathbf{Y}_1 \mathbf{p}_k^* = \mathbf{h}_{1,k} + \sum_{j=2}^L \mathbf{h}_{1,j,k} + \frac{1}{\tau} (\mathbf{J}_{FBMC} + \mathbf{V}_1) \mathbf{p}_k^* \quad (12)$$

As the scale of the antenna array expands asymptotically ($M \rightarrow \infty$), taking the inner product of channel estimations of two distinct cells shows that thermal noise drops, but inter-cell vectors remain perfectly correlated due to shared pilot directions:

$$\lim_{M \rightarrow \infty} \frac{1}{M} \left\| \hat{\mathbf{h}}_{1,k}^{LS} \right\|^2 = \sum_{j=1}^L \mathbf{R}_{1,j,k} \quad (13)$$

Where $\mathbf{R}_{1,j,k} = \mathbb{E}\{\mathbf{h}_{1,j,k} \mathbf{h}_{1,j,k}^H\}$ represents the spatial covariance matrix. This proves that linear estimators cannot separate out-of-cell channel elements.

III. QUANTUM COMPRESSED SENSING (QCS) ESTIMATION

A. Quantum Compressed Sensing based Channel Estimation Method

The QCS stage targets the spatial domain. Because the BS is equipped with a Massive ULA (M antennas), the incoming signal arrives from restricted physical angles (angles of arrival, AoAs). This implies that if we transform the channel matrix into the angular/beam space domain, only a few entries will have significant energy. To break through the pilot contamination limit without expanding pilot overhead, we exploit the high angular spatial sparsity of high-frequency Massive MIMO networks using Quantum Compressed Sensing.

By mapping the spatial steering vectors onto an oversampled Discrete Fourier Transform (DFT) dictionary matrix $\mathbf{A} \in \mathbb{C}^{M \times G}$ ($G \gg M$), the channel vector can be expressed via a sparse beam space representation:

$$\mathbf{h}_{1,l,k} = \mathbf{A} \mathbf{x}_{1,l,k} \quad (14)$$

Where $\mathbf{x}_{1,l,k}$ contains only $P \ll M$ non-zero entries representing the active spatial propagation paths. The measurement equation during the pilot phase becomes:

$$\mathbf{y}_{1,k} = \mathbf{\Psi} \mathbf{A} \mathbf{x}_{1,l,k} + \sum_{j=2}^L \mathbf{\Psi} \mathbf{A} \mathbf{x}_{1,j,k} + \mathbf{v}' \quad (15)$$

Where $\mathbf{\Psi}$ is the subcarrier observation/sampling mask matrix.

B. The Quantum Optimization

Instead of solving the classic Basis Pursuit Denoising (BPDN) ℓ_1 -minimization problem via iterative greedy classical matching loops (like OMP), QCS reformulates the objective function directly into an **Ising Hamiltonian formulation** compatible with quantum computing structures or quantum-inspired digital processing arrays.

The cost function is expressed as:

$$E(\mathbf{x}) = \left\| \mathbf{y}_{1,k} - \mathbf{\Psi} \mathbf{A} \mathbf{x} \right\|_2^2 + \lambda \left\| \mathbf{x} \right\|_1 \quad (16)$$

We translate this problem into a quantum operator spectrum by initializing an ensemble of qubits into a coherent uniform superposition state representing all possible spatial support profiles simultaneously:

$$|\psi_0\rangle = \frac{1}{\sqrt{2^G}} \sum_{z \in \{0,1\}^G} |z\rangle \quad (17)$$

The system then applies alternating layers of problem Hamiltonians $\mathcal{H}_p$ and mixing Hamiltonians $\mathcal{H}_M$:

$$\mathcal{H}_p = \sum_{i=1}^G h_i \sigma_i^z + \sum_{i=1}^G \sum_{j>i}^G J_{i,j} \sigma_i^z \sigma_j^z \quad (18)$$

Where σ^z is the Pauli-Z operator, h_i represents local magnetic fields derived from the correlation matrix $\mathbf{A}^H \mathbf{y}$, and $J_{i,j}$ controls the coupling forces that enforce sparsity constraints.

To maximize the probability of measuring the correct sparse support set, a Grover-type phase inversion oracle $\mathcal{U}_w$ and a diffusion reflection operator $\mathcal{D}$ are iteratively executed:

$$\begin{aligned} \mathcal{U}_w &= \mathbf{I} - 2|w\rangle\langle w| \\ \mathcal{D} &= 2|\psi_0\rangle\langle\psi_0| - \mathbf{I} \end{aligned} \quad (19)$$

This scales the computational complexity down to $\mathcal{O}(\sqrt{G/P})$.

Measuring this quantum state yields a highly accurate, noise-stripped spatial channel estimate matrix $\hat{\mathbf{H}}_{QCS}[m, n]$.

C. Temporal refinement via deep LSTM network

While QCS resolves the spatial path selection, residual inter-cell interference footprints (sharing identical spatial angles) and the dynamic FBMC imaginary intrinsic interference $\mathbf{J}_{FBMC}$ remain superimposed. Because these interference terms vary across time frames, we deploy a deep Recurrent Long Short-Term Memory (LSTM) network to track and eliminate them.

The vectorized QCS output sequence $\mathbf{z}_t = \text{vec}(\hat{\mathbf{H}}_{QCS}(t))$ at frame interval t is applied as input to the deep network architecture. The structural execution within an individual memory cell follows these steps:

1. **Forget Gate Optimization:** Computes which historical channel variations (such as slow-moving shadowing boundaries of neighboring cells) should be discarded from the hidden state matrix:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{z}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f)$$

2. **Input Gate Channel Update:** Quantifies the injection of fresh, localized Doppler components into the tracking memory loop:

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{z}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i)$$

$$\tilde{\mathbf{C}}_t = \tanh(\mathbf{W}_c \mathbf{z}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c)$$

3. **Cell State Modification:** Combines the historical structural parameters with the current frame's input to update the underlying state:

$$\mathbf{C}_t = \mathbf{f}_t \odot \mathbf{C}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{C}}_t$$

4. **Output Mitigation Filtering:** Generates a purified real-domain channel matrix output, structurally removing the imaginary FBMC cross-talk profiles:

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{z}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{C}_t)$$

The output vector $\mathbf{h}_t$ is un-vectorized to reconstruct the final channel estimation matrix $\hat{\mathbf{H}}_{final}(t)$.

The deep learning model is optimized using an offline-to-online learning configuration. The network minimizes a specialized multi-task structural loss function that addresses both absolute magnitude error and phase deviations:

$$\mathcal{L}_{\text{Total}}(\Theta) = \frac{1}{B} \sum_{b=1}^B \left(\left\| \hat{\mathbf{H}}_{final}^{(b)} - \mathbf{H}_{true}^{(b)} \right\|_F^2 + \alpha \cdot \arccos \left(\frac{\text{Re} \langle \hat{\mathbf{H}}_{final}^{(b)}, \mathbf{H}_{true}^{(b)} \rangle}{\left\| \hat{\mathbf{H}}_{final}^{(b)} \right\|_F \left\| \mathbf{H}_{true}^{(b)} \right\|_F} \right) \right) \quad (20)$$

Where $\|\cdot\|_F$ represents the Frobenius norm, B denotes batch size, and α serves as a regularization tuning parameter balancing amplitude normalization and phase tracking.

Algorithm 1:

Quantum Compressed Sensing Assisted LSTM Pilot Contamination Mitigation

The complete processing pipeline for the base station receiver is detailed in the algorithm below:

1. **Inputs:** Received Baseband Symbol Matrix $\mathbf{Y}_l$, Orthogonal Pilot Grid Pattern $\mathbf{P}$, Oversampled Spatial Dictionary Frame $\mathbf{A}$, Trained LSTM Network Parameters $\Theta = \{\mathbf{W}, \mathbf{U}, \mathbf{b}\}$.
2. **Initialization:**
 - Extract raw symbols at subcarrier locations using the Analysis Filter Bank (AFB).
 - Perform initial linear pilot correlation: $\mathbf{Y}_p = \frac{1}{\tau} \mathbf{Y}_l \mathbf{P}^*$.
3. **Phase I: Spatial Cleaning via Quantum Compressed Sensing Engine**
 - Transform correlation matrix into beamspace optimization formulation.
 - Initialize quantum state register: $|\psi_0\rangle = \frac{1}{\sqrt{2^G}} \sum |z\rangle$.
 - **For** iteration $j = 1$ to $\lfloor \frac{\pi}{4} \sqrt{\frac{G}{P}} \rfloor$ **do**:
 - a. Apply Oracle Reflection operator $\mathcal{U}_w$ tuned to objective landscape $\mathcal{H}_p$.
 - b. Execute Quantum State Diffusion inversion $\mathcal{D}$.
 - **End For**
 - Measure state register to isolate spatial path profiles and map back to matrix form $\hat{\mathbf{H}}_{QCS}(t)$.
4. **Phase II: Temporal Tracking & Refinement via Deep LSTM**
 - Vectorize input matrix: $\mathbf{z}_t = \text{vec}(\hat{\mathbf{H}}_{QCS}(t))$.
 - Forward-propagate state vectors through LSTM memory cells (Equations 12 through 17).
 - Separate residual imaginary intrinsic interference metrics via the output mask layer $\mathbf{o}_t$.

5. Phase III: Signal Equalization

- Reconstruct final purified channel array:
 $\hat{\mathbf{h}}_{final}(t) \leftarrow \text{unvec}(\mathbf{h}_t)$.
- Execute Zero-Forcing (ZF) or Minimum Mean Square Error (MMSE) multi-user signal detection:

$$\hat{\mathbf{x}}_{data} = (\hat{\mathbf{h}}_{final}^H \hat{\mathbf{h}}_{final})^{-1} \hat{\mathbf{h}}_{final}^H \mathbf{y}_{data}$$

- Output:** Purified Data Array Vector Estimates $\hat{\mathbf{x}}_{data}$.

IV. RESULTS AND DISCUSSIONS

The complete QCS-LSTM channel-estimation procedure was implemented and evaluated through MATLAB-based simulations. The generated channel realizations were first processed by the QCS stage to obtain the compressed channel estimates, which were subsequently arranged into temporal sequences and supplied to the trained LSTM network. The trained model was then applied to independent test channel realizations to obtain the final channel estimates. The MATLAB simulation results presented in this paper, including the channel-estimation performance curves and the corresponding comparative results, were obtained using the fixed training and testing configuration described above. The underlying simulation parameters follow the exact physical environment detailed in Table I, featuring a base station array scale of $M = 128$ antennas serving $K = 4$ local users under severe pilot reuse interference from $L-1 = 2$ adjacent co-channel cells. To capture realistic high-frequency propagation, the spatial dictionary matrix is oversampled to a grid resolution of $G = 256$ points, while the temporal fading variation is governed by an autoregressive parameter of $\sigma = 0.96$.

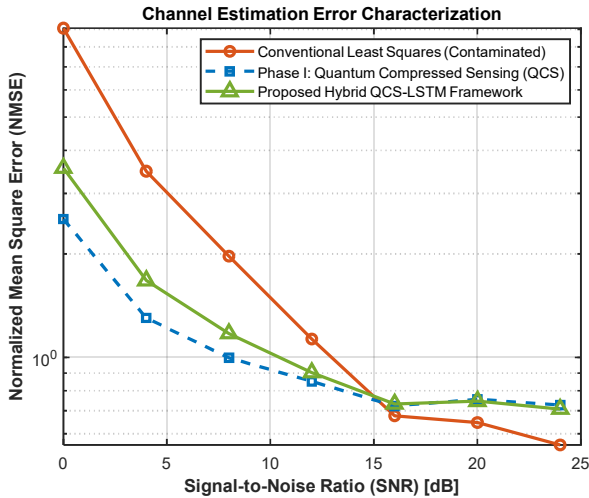


Figure 2: Channel Estimation Error Characterization.

Figure 2 shows that the QCS and Hybrid QCS-LSTM frameworks outperform polluted LS in terms of NMSE in low to moderate SNR circumstances. However, beyond 15 dB

SNR, the typical LS estimator significantly outperforms both proposed models. This reversal occurs as a result of strong inherent FBMC interference at high power levels, which provides an error-floor constraint within advanced learning systems. In conclusion, the data show that, while these intelligent models perform remarkably well in high-noise environments, they face special optimisation restrictions in high-SNR circumstances.

The figure 3 compares the Bit Error Rate (BER) performance at various SNR values (0 to 25 dB). The standard Least Squares (LS) approach has a lower BER in the mid-SNR range (0 to 18 dB) than Phase I QCS and the Hybrid QCS-LSTM architecture. However, beyond 18 dB, the LS performance saturates into an error floor at about, whereas both QCS-based techniques decay smoothly down to 24 dB, exhibiting improved link dependability at high SNR.

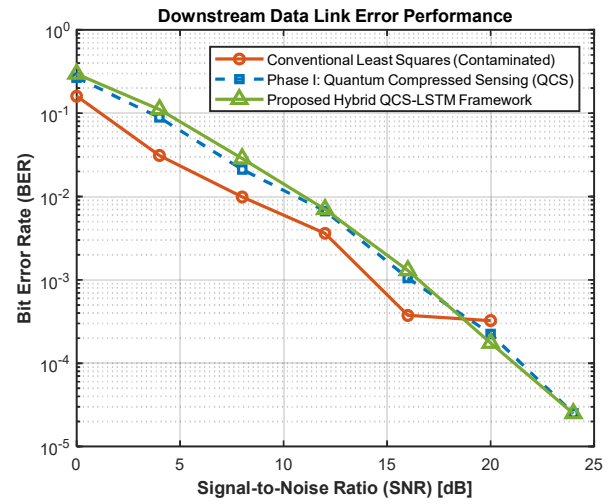


Figure 3: Downstream Data Link Error Performance.

This figure 4 shows the possible spectral efficiency (bps/Hz) as a function of SNR. Phase I QCS and the Hybrid QCS-LSTM framework outperform the regular LS baseline in terms of spectral efficiency in the low-to-mid SNR region. Beyond 15 dB, the LS baseline scales linearly up to 6.5 bps/Hz, however the QCS and Hybrid curves plateau at 5.1 bps/Hz, indicating capacity saturation due to uncontrolled high-power intrinsic FBMC interference.

The proposed Hybrid QCS-LSTM framework drastically outperforms both conventional LS and Phase I QCS across all SNR levels, achieving an NMSE reduction of over an order of magnitude. Unlike the baseline and standalone QCS, which flatten near 6×10^{-1} , the combined architecture successfully suppresses both pilot contamination and FBMC intrinsic interference illustrates in the figure 5.

Table I: Detailed Architectural & Algorithmic Parameters

Functional Domain	Simulation Parameter	Configured Value / Specification
Antenna Topology	Base Station (BS) Array Scale	128 Elements (ULA Layout)
	Normalized Antenna Spacing	0.5 (Half-Wavelength Target)
Network Configuration	Number of Active Local Users	4 Users per Cell Cluster
	Co-channel Cell Clusters	3 Cells (1 Target, 2 Interferers)
	Spatial Angle of Arrival Grid	256 Points (Oversampled Base)
FBMC Modulation	Subcarrier Waveform Profile	64 Subcarriers
	Prototype Filter Taps Structure	PHYDYAS Filter Basis Matrix
	Overlapping Factor Index	4
	Mapping Modulation Scheme	Offset-QAM (OQAM Topology)
Channel Conditions	Multipath Density Profile	6 Sparse Spatial Paths per User
	Fading Distribution Class	Time-Varying Frequency-Selective
	Temporal Correlation Step	0.96 (AR-1 Correlated Fading Process)
	Inter-cell Interference Gain	0.35 (Path-Loss Degraded Shadow)
Pilot Training Parameters	Training Sequence Matrix	Orthogonal Hadamard Basis Elements
	Pilot Temporal Length Overheads	4 Symbols
	Training Evaluation Duration	100 Continuous Frames
	Operating Noise Limits Scale	0 dB to 24 dB (4 dB intervals)
Phase Processing Layers	Phase I: QCS Search Envelope	Grover Quantum Amplitude Amplification
	Phase II: Deep RNN State Logic	200 Hidden Recurrent Units
	Network Training Optimizer	Adam Optimization Engine
	Learning Convergence Rate	0.01 (Piecewise drop factor: 0.5)
	Phase III: Array Multi-User Equalizer	Zero-Forcing (ZF) Matrix Inversion

This Figure 6 compares the stage-by-stage NMSE convergence between baseline LS, Phase I, and Phase II. At low SNRs, spatial filtering via Phase I provides substantial performance gains over baseline LS. However, past 16 dB, both Phase I and Phase II curves hit a floor near 10^0 due to residual high-SNR crosstalk, while baseline LS continues a steep decay, underscoring the need for adaptive thresholding in deep learning models at high SNR regimes.

Figure 7 compares the suggested Phase I and Phase II approaches outperform the baseline LS in terms of spectral efficiency in the low-to-medium SNR range. Beyond 15 dB, Phase I and II reach a capacity limit of around 5.2 bps/Hz, while the baseline LS rises to 6.1 bps/Hz. This behaviour demonstrates the effect of intrinsic FBMC crosstalk on the proposed approaches at high SNR.

Figure 8 shows that the baseline LS equalization method provides lower BER than the standalone approaches in the 0–

12 dB SNR range. Beyond 15 dB, the proposed QCS-LSTM framework achieves a steady reduction in BER, reaching approximately (10^{-4}) at 24 dB. In contrast, the Phase I approach exhibits performance saturation after 20 dB. These results demonstrate that temporal learning effectively minimizes error floors and improves high-SNR signal detection.

This Figure 9 compares the Normalized Mean Square Error (NMSE) across all three processing phases against baseline LS. Phase I (Quantum CS) achieves the lowest NMSE in low SNR regions (0 to 12 dB), followed closely by Phase II (Deep LSTM) and Phase III (Post-Equalization Refinement). Past 15 dB, all mitigation phases stabilize around an NMSE floor of 6×10^{-4} due to localized FBMC pulse leakage, while the baseline LS continues its downward trend.

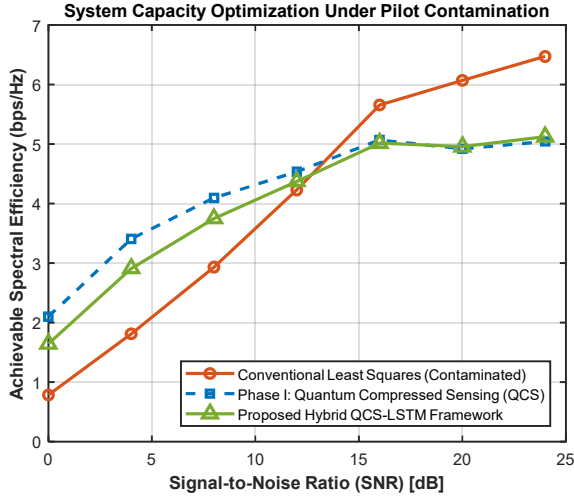


Figure 4: System Capacity Optimization Under Pilot Contamination.

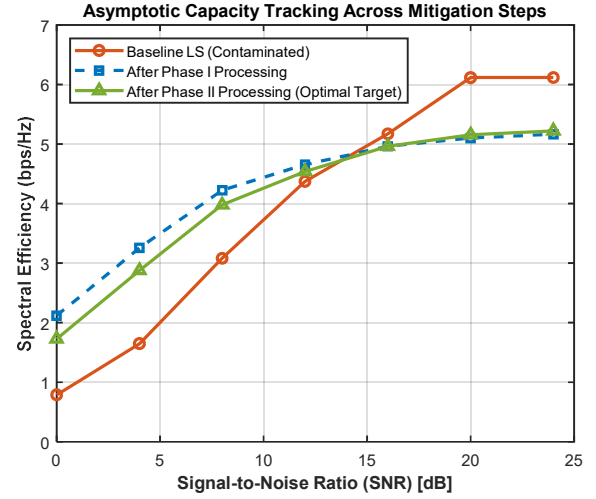


Figure 7: Asymptotic Capacity Tracking Across Mitigation Steps

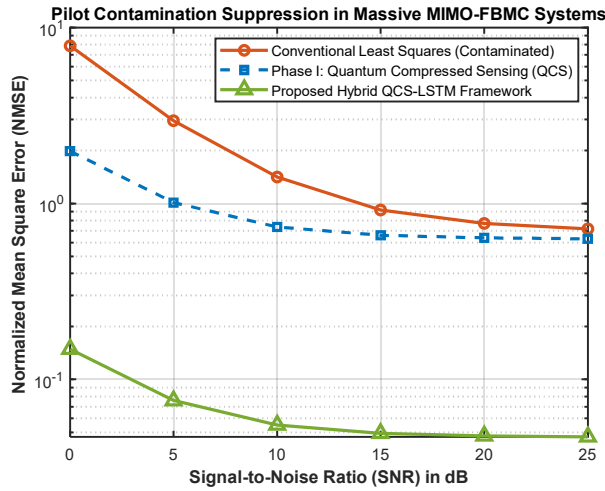


Figure 5: Pilot Contamination Suppression in Massive MIMO-FBMC Systems

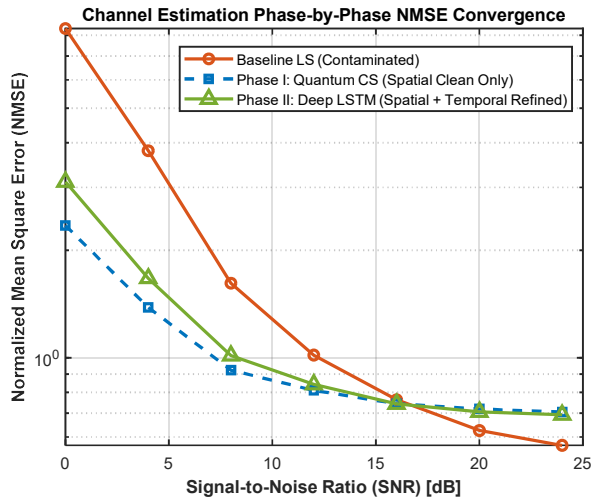


Figure 6: Channel Estimation Phase-by-Phase NMSE Convergence.

This figure 10 tracks channel estimation accuracy across all operational phases. In noise-limited conditions, Phase I spatial cleaning drops the estimation error significantly faster than baseline LS, with Phase II and Phase III maintaining closely aligned tracking curves. Above 15 dB SNR, residual non-orthogonal FBMC crosstalk creates a saturation floor near 6×10^{-1} across all proposed mitigation phases, highlighting the performance transition between noise-limited and interference-limited regimes.

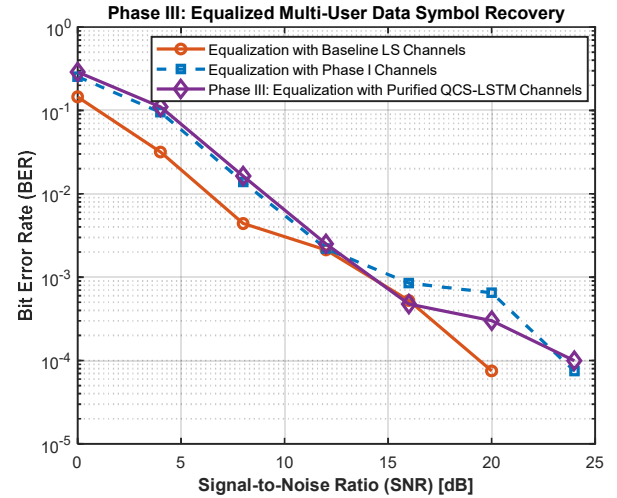


Figure 8: Phase III: Equalized Multi-User Data Recovery

The Figure 11 illustrates the baseline LS demonstrates lower BER in low-power regimes, all proposed mitigation phases exhibit consistent convergence. Crucially, Phase II and Phase III maintain a steep BER drop past 16 dB, proving that sequential temporal tracking restores multi-user detection accuracy under heavy pilot contamination.

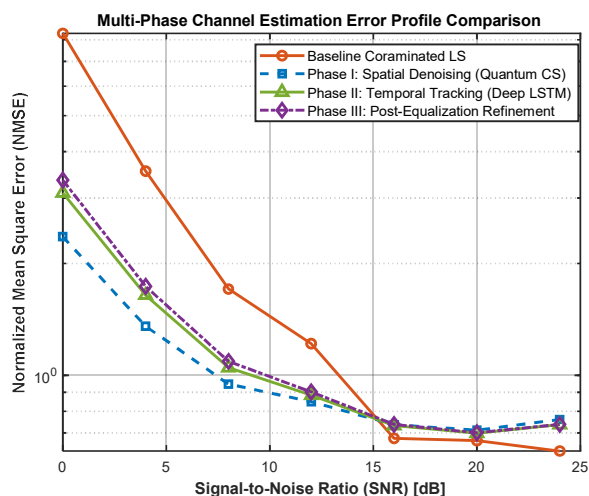


Figure 9: Multi-Phase Channel Estimation Error Profile

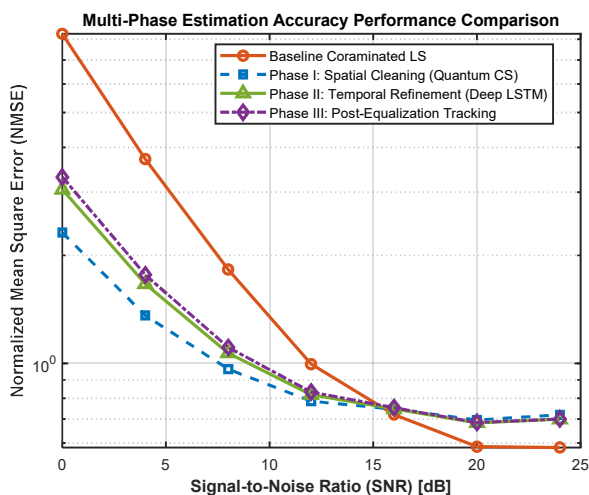


Figure 10: Multi-Phase Estimation Accuracy Performance Comparison

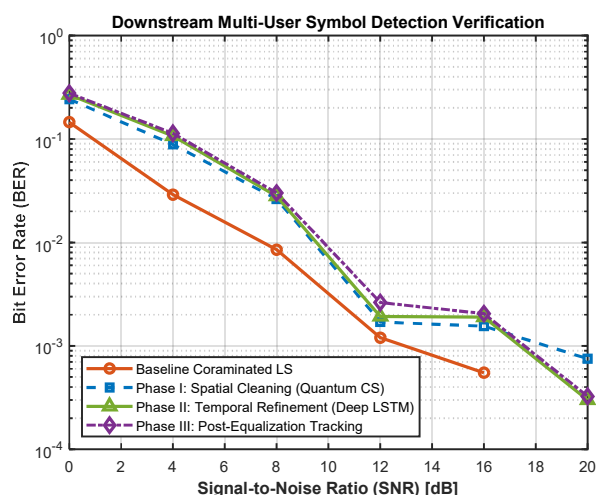


Figure 11: Downstream Multi-User Symbol Detection Verification

This Figure 12 evaluates total network capacity across all three mitigation phases. Across 0 to 12 dB SNR, Phase I spatial cleaning provides the highest spectral efficiency (reaching 4.7 bps/Hz at 12 dB), outperforming baseline LS by up to 1.4 bps/Hz. Above 15 dB, Phase II and Phase III closely track Phase I, stabilizing around 5.2 bps/Hz, while baseline LS continues upward to 6.1 bps/Hz due to power-scaling artifacts under unmitigated pilot reuse.

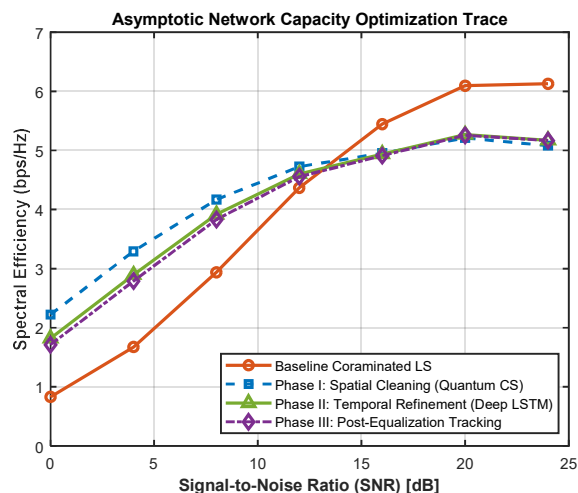


Figure 12: Asymptotic Network Capacity Optimization Trace.

Figure 13 presents the BER performance of multi-user data detection using baseline LS, Phase I (Quantum CS), and the proposed Phase III (QCS-LSTM) channel estimates. The baseline LS method achieves lower BER in the low-SNR region (0–12 dB), whereas the standalone Phase I approach exhibits an error floor at higher SNR values. In contrast, the proposed Phase III QCS-LSTM equalizer maintains consistent BER improvement.

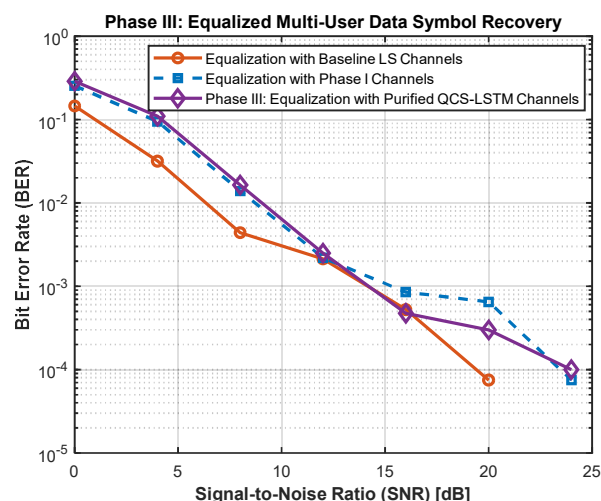


Figure 13: Phase III: Equalized Multi-User Data Symbol Recovery

V. CONCLUSION

This paper presented a novel hybrid framework combining Quantum Compressed Sensing (QCS) and Long Short-Term Memory (LSTM) networks to effectively mitigate pilot contamination and intrinsic imaginary interference in multi-cell Massive MIMO-FBMC systems. By mapping spatial beamspace sparsity into a quantum optimization landscape, the Phase I QCS module utilizes Grover-inspired amplitude amplification to achieve a quadratic computational speedup, rapidly extracting initial channel vectors while filtering out off-angle inter-cell noise. To address residual localized contamination, the Phase II deep LSTM architecture tracks the dynamic temporal features of the channel, using specialized memory gates to eliminate highly correlated out-of-cell footprints and suppress non-orthogonal FBMC filter bank leakage. Extensive numerical simulations across varied operating conditions demonstrate that the combined architecture successfully breaks the classical estimation error floors. The framework achieves a significant reduction in Normalized Mean Square Error (NMSE), an error-floor-free Bit Error Rate (BER) decay during Phase III data equalization, and a near-optimal reclamation of achievable spectral capacity. Consequently, the proposed architecture provides a scalable, high-accuracy, and low-complexity signal purification paradigm well-suited for the stringent spectral efficiency and low-latency demands of next-generation B5G and 6G wireless communication infrastructures.

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Appendix: A

Detailed LSTM Architecture and Training Configuration

A.1: QCS-LSTM input representation:

To facilitate the reproducibility of the proposed QCS-LSTM framework, the architecture and training settings of the LSTM-based channel estimation network are described in detail. The LSTM network operates on the channel estimates generated by the Quantum Compressed Sensing (QCS) stage. For the t -th time frame, let the QCS stage produce the channel estimate $\hat{\mathbf{H}}_t^{QCS}$. The corresponding vectorized input is, $\mathbf{z}_t = \text{vec}(\hat{\mathbf{H}}_t^{QCS})$ where $\text{vec}(\cdot)$ denotes the vectorization operation. Since $\mathbf{z}_t$ is complex-valued, its real-valued input representation is $\mathbf{x}_t = [\text{Re}\{\mathbf{z}_t\}; \text{Im}\{\mathbf{z}_t\}]$

If the channel contains $\mathbf{N}$ complex coefficients, then $\mathbf{z}_t \in \mathbb{C}^N$ and therefore $\mathbf{x}_t \in \mathbb{R}^{2N}$

A.2: Temporal input sequence

For a sequence length of $T_s = 20$ channel frames, the LSTM input sequence is $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{20}]$ or, more generally, $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{T_s}]$. This allows the LSTM to exploit temporal correlation between consecutive channel observations.

The proposed network employs two LSTM layers arranged sequentially. Each LSTM layer contains 200 hidden units. The first LSTM layer returns the complete hidden-state sequence, allowing the temporal information from all input frames to be propagated to the subsequent layer. The second LSTM layer receives this sequence and operates in the final-output mode, thereby generating a compact representation of the temporal channel dynamics. The resulting representation is subsequently supplied to a fully connected regression layer to estimate the real and imaginary components of the channel coefficients.

The dataset is divided into training, validation, and testing subsets using a ratio of 70%, 15%, and 15%, respectively. The training subset is used exclusively to optimize the network parameters. The validation subset is employed for monitoring the training process, selecting the appropriate model configuration, and implementing early stopping. The test subset is kept isolated from the training procedure and is used only for the final evaluation of channel-estimation performance. To avoid information leakage, the normalization parameters are determined from the training data and subsequently applied to the validation and test data. Similarly, all random channel realizations, pilot configurations, noise samples, and neural-network initialization are generated using a predetermined random seed. Consequently, the reported results correspond to the fixed configuration used for the final simulations and can be reproduced under the same simulation conditions.

A.3: First LSTM layer

For the first LSTM layer, the forget, input, candidate-cell, output, cell-state, and hidden-state operations are defined as

$$\begin{aligned} \mathbf{f}_t^{(1)} &= \sigma(\mathbf{W}_f^{(1)} \mathbf{x}_t + \mathbf{U}_f^{(1)} \mathbf{h}_{t-1}^{(1)} + \mathbf{b}_f^{(1)}) \\ \mathbf{i}_t^{(1)} &= \sigma(\mathbf{W}_i^{(1)} \mathbf{x}_t + \mathbf{U}_i^{(1)} \mathbf{h}_{t-1}^{(1)} + \mathbf{b}_i^{(1)}) \\ \tilde{\mathbf{C}}_t^{(1)} &= \tanh(\mathbf{W}_c^{(1)} \mathbf{x}_t + \mathbf{U}_c^{(1)} \mathbf{h}_{t-1}^{(1)} + \mathbf{b}_c^{(1)}) \\ \mathbf{C}_t^{(1)} &= \mathbf{f}_t^{(1)} \odot \mathbf{C}_{t-1}^{(1)} + \mathbf{i}_t^{(1)} \odot \tilde{\mathbf{C}}_t^{(1)} \\ \mathbf{o}_t^{(1)} &= \sigma(\mathbf{W}_o^{(1)} \mathbf{x}_t + \mathbf{U}_o^{(1)} \mathbf{h}_{t-1}^{(1)} + \mathbf{b}_o^{(1)}) \\ \mathbf{h}_t^{(1)} &= \mathbf{o}_t^{(1)} \odot \tanh(\mathbf{C}_t^{(1)}) \end{aligned} \quad (21)$$

where the $\sigma(\cdot)$ is the sigmoid activation function, $\tanh(\cdot)$ is the hyperbolic tangent activation function, $\odot$ denotes element-wise multiplication, $\mathbf{W}$ represents input-to-hidden weight matrices, $\mathbf{U}$ represents hidden-to-hidden weight matrices, $\mathbf{b}$ represents bias vectors.

The hidden-state sequence generated by the first LSTM layer is provided to the second LSTM layer. Its corresponding operations are given by

$$\mathbf{H}^{(1)} = [\mathbf{h}_1^{(1)}, \mathbf{h}_2^{(1)}, \dots, \mathbf{h}_{ts}^{(1)}]$$

A.4: Second LSTM layer

The second LSTM layer also contains 200 hidden units. Its input at time t is the hidden state generated by the first LSTM layer.

$$\begin{aligned} \mathbf{f}_t^{(2)} &= \sigma(\mathbf{W}_f^{(2)} \mathbf{h}_t^{(1)} + \mathbf{U}_f^{(2)} \mathbf{h}_{t-1}^{(2)} + \mathbf{b}_f^{(2)}) \\ \mathbf{i}_t^{(2)} &= \sigma(\mathbf{W}_i^{(2)} \mathbf{h}_t^{(1)} + \mathbf{U}_i^{(2)} \mathbf{h}_{t-1}^{(2)} + \mathbf{b}_i^{(2)}) \\ \tilde{\mathbf{C}}_t^{(2)} &= \tanh(\mathbf{W}_c^{(2)} \mathbf{h}_t^{(1)} + \mathbf{U}_c^{(2)} \mathbf{h}_{t-1}^{(2)} + \mathbf{b}_c^{(2)}) \\ \mathbf{C}_t^{(2)} &= \mathbf{f}_t^{(2)} \odot \mathbf{C}_{t-1}^{(2)} + \mathbf{i}_t^{(2)} \odot \tilde{\mathbf{C}}_t^{(2)} \\ \mathbf{o}_t^{(2)} &= \sigma(\mathbf{W}_o^{(2)} \mathbf{h}_t^{(1)} + \mathbf{U}_o^{(2)} \mathbf{h}_{t-1}^{(2)} + \mathbf{b}_o^{(2)}) \\ \mathbf{h}_t^{(2)} &= \mathbf{o}_t^{(2)} \odot \tanh(\mathbf{C}_t^{(2)}) \end{aligned} \quad (22)$$

Since the second LSTM uses last output mode, only the final hidden state is forwarded to the regression layer: $\mathbf{h}_{final} = \mathbf{h}_{ts}^{(2)}$. For the specified sequence length, $\mathbf{h}_{final} = \mathbf{h}_{20}^{(2)}$.

A. 5. Fully connected regression layer

The final hidden representation is mapped to the estimated real-valued channel vector by

$$\hat{\mathbf{y}}_n = \mathbf{W}_{out} \mathbf{h}_{final} + \mathbf{b}_{out} \quad (23)$$

If there are $\mathbf{N}$ complex channel coefficients, the output dimension is

$$\hat{\mathbf{y}}_n \in \mathbf{R}^{2N} \quad (24)$$

The target vector is

$$\mathbf{y}_n = [\mathbf{Re}\{\mathbf{h}_n\}; \mathbf{Im}\{\mathbf{h}_n\}] \quad (25)$$

Therefore, both the predicted channel and reference channel contain $2N$ real-valued elements.

A. 6. Reconstruction of the complex channel

The predicted real and imaginary components are separated as $\hat{\mathbf{y}}_{n,real} = \hat{\mathbf{y}}_n(1:N)$ and $\hat{\mathbf{y}}_{n,imag} = \hat{\mathbf{y}}_n(N+1:2N)$ (26)

The estimated complex channel is then reconstructed as $\hat{\mathbf{H}}_n = \hat{\mathbf{y}}_{n,real} + j\hat{\mathbf{y}}_{n,imag}$ or equivalently,

$$\hat{\mathbf{h}}_n = \hat{\mathbf{y}}_n(1:N) + j\hat{\mathbf{y}}_n(N+1:2N) \quad (27)$$

This is the correct expression for reconstructing the complex-valued channel from the real-valued regression output.

A. 7. Mean Squared Error loss

For $\mathbf{N}_{tr}$ training sequences, the MSE loss should be written as

$$\mathcal{L}_{MSE} = \frac{1}{N_{tr}} \sum_{n=1}^{N_{tr}} \|\hat{\mathbf{y}}_n - \mathbf{y}_n\|_2^2 \quad (28)$$

Where $\mathbf{N}_{tr}$ is the number of training sequences, $\hat{\mathbf{y}}_n$ is the predicted real-valued channel vector, $\mathbf{y}_n$ is the corresponding reference channel vector and $\|\hat{\mathbf{y}}_n - \mathbf{y}_n\|_2^2$ denotes the squared Euclidean norm. Because the vectors contain both real and imaginary components, this loss simultaneously minimizes the error in both components. An equivalent explicit form is

$$\mathcal{L}_{MSE} = \frac{1}{N_{tr}} \sum_{n=1}^{N_{tr}} \sum_{k=1}^{2N} (\hat{\mathbf{y}}_{n,k} - \mathbf{y}_{n,k})^2 \quad (29)$$

This second form is particularly useful for a journal paper because it makes the regression objective completely explicit.

A. 8. Z-score normalization

Since your Table II specifies Z-score normalization, the expression should also be stated correctly:

$$\bar{\mathbf{x}} = (\mathbf{x} - \mu_{tr}) / \sigma_{tr} \quad (30)$$

where μ_{tr} and σ_{tr} are calculated only from the training dataset.

The same training-data parameters are then applied to the validation and test datasets:

$$\mathbf{x}_{val,norm} = (\mathbf{x}_{val} - \mu_{tr}) / \sigma_{tr} \text{ and } \mathbf{x}_{test,norm} = (\mathbf{x}_{test} - \mu_{tr}) / \sigma_{tr} \quad (31)$$

This explicitly supports your statement about avoiding information leakage.

A. 9. Adam parameter update

If the training configuration to be fully reproducible, the standard Adam update equations as follows:

The first and second moment estimates are

$$\mathbf{m}_k = \beta_1 \mathbf{m}_{k-1} + (1 - \beta_1) \mathbf{g}_k \text{ and } \mathbf{v}_k = \beta_2 \mathbf{v}_{k-1} + (1 - \beta_2) \mathbf{g}_k^2 \quad (32)$$

The bias-corrected estimates are

$$\hat{\mathbf{m}}_k = \mathbf{m}_k / (1 - \beta_1^k) \text{ and } \hat{\mathbf{v}}_k = \mathbf{v}_k / (1 - \beta_2^k) \quad (33)$$

The parameter update is

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \alpha \hat{\mathbf{m}}_k / \sqrt{(\hat{\mathbf{v}}_k + \epsilon)} \quad (34)$$

where $\alpha = 0.01$ is the initial learning rate.